Lec 35

Convergence of RVs.

(a) almost-sure convergence $X_1, X_2, \dots$ are defined over $(\Omega, \mathcal{F}, P)$.

$X_n \xrightarrow[a.s]{} X$ if

$$P\left(\left\{\omega : \lim_{n\to\infty} X_n(\omega) = X(\omega)\right\}\right) = 1$$

Examples:

$P(\boxed{[0.1)}) = 1$

1. $X_n(\omega) = \omega^n$, $X(\omega) = 0$. $X_n \xrightarrow[a.s]{} X$.
2. $S_n = \frac{\sum_{i=1}^{n} X_i}{n}$ $X = E[X_1]$. X_i's are iid. $S_n \xrightarrow[a.s]{} \mu$.
3. $Y_n = \min(X_1, \dots X_n)$ X_i's iid uniform.

$Y_n \xrightarrow[a.s]{} 0$.

(b) Convergence in probability

$X_n \xrightarrow[probability]{} X$ if $\lim_{n\to\infty} P(|X_n - X| > \varepsilon) = 0$.

1. Moving-shrinking rectangles. 2. $S_n \xrightarrow{probability} \mu$

$$A_n = \{\omega : |X_n(\omega) - X(\omega)| > \varepsilon\}$$

$$\limsup A_n = \bigcap_{N=1}^{\infty} \bigcup_{n \geq N} A_n$$

Equivalent condition for a.s convergence

is $P(\limsup A_n) = 0$.

Borel-Cantelli Lemma gives a sufficient condition for $P(\limsup A_n) = 0$ to be satisfied.

$$\boxed{\sum_{n=1}^{\infty} P(A_n) < \infty} \Longrightarrow P(\limsup A_n) = 0$$

Strong law of large Numbers.

$X_1, X_2, \dots$ are iid with finite mean μ and $E[\bar{X}_1^4] < \infty$.

and let $S_n = \dfrac{\sum_{i=1}^{n} X_i}{n}$ $\qquad S_n \xrightarrow[a.s]{} \mu$

We'll take help from Borel Cantelli Lemma.

$$A_n = \{\omega : |S_n(\omega) - \mu| > \varepsilon\} = \{\omega : |\bar{S}_n(\omega)| > \varepsilon\}$$

$$\begin{aligned}
\bar{S}_n &= S_n - \mu \\
&= \frac{\sum_{i=1}^{n} X_i}{n} - \mu = \frac{\sum_{i=1}^{n} (X_i - \mu)}{n} \\
&= \frac{\sum_{i=1}^{n} \bar{X}_i}{n}
\end{aligned}$$

$$P(A_n) = P(|\bar{S}_n| > \varepsilon)$$
$$= P(\bar{S}_n^4 > \varepsilon^4)$$
$$\le \frac{E[\bar{S}_n^4]}{\varepsilon^4}$$

$$E[\bar{S}_n^4] = \frac{E\left[\left(\sum_{i=1}^{n} \bar{X}_i\right)^4\right]}{n^4}$$

$$(\bar{X}_1 + \cdots \bar{X}_n)(\quad)$$
$$(\quad)(\quad)$$

$$\bar{X}_1^4 \quad \bar{X}_1^2\bar{X}_2\bar{X}_3 \quad \bar{X}_1\bar{X}_2\bar{X}_3\bar{X}_4 \quad \bar{X}_1^3\bar{X}_2$$
$$\bar{X}_1^2\bar{X}_2^2$$

$$\sum_{n=1}^{\infty} \frac{1}{n^p}$$

Converges $p>1$

diverges otherwise

$$E[\bar{X}_1^2\bar{X}_2\bar{X}_3] = E[\bar{X}_1^2]\,\underbrace{E[\bar{X}_2]}_{=0}\,\underbrace{E[\bar{X}_3]}_{=0} = 0$$

$$E[\bar{S}_n^4] = \frac{n\,E[\bar{X}_1^4] + \binom{n}{2}\binom{4}{2}\left(E[\bar{X}_1^2]\right)^2}{n^4} \qquad \left(E[\bar{X}_1^2]\right)^2 = E[\bar{X}_1^2\bar{X}_2^2]$$

$$\left(E[\bar{X}_1^2]\right)^2 \le E[\bar{X}_1^4] \iff \mathrm{Var}(\bar{X}_1^2) \ge 0$$
$$\mathrm{Var}(\bar{X}_1^2) = E[\bar{X}_1^4] - \left(E[\bar{X}_1^2]\right)^2$$

$$E[\bar{S}_n^4] \le \frac{E[\bar{X}_1^4]}{n^4}\left[n + \frac{n(n-1)}{\cancel{2}}\cancel{6}^{\,3}\right] \le \frac{3\cancel{n^2}}{\cancel{n^4}\,n^2}E[\bar{X}_1^4]$$

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Convergence of RVs

$$E[\bar{S}_n^4] \leq \frac{3}{n^2} E[X_1^4].$$

$$P(A_n) = P(|\bar{S}_n| > \varepsilon) \leq \frac{E[\bar{S}_n^4]}{\varepsilon^4} \leq \frac{3}{n^2\varepsilon^4} E[X_1^4].$$

$$\therefore \sum_{n=1}^{\infty} P(A_n) \leq \left[\sum_{n=1}^{\infty} \frac{1}{n^2}\right] \frac{3}{\varepsilon^4} E[X_1^4].$$

$$< \infty.$$

$$\Rightarrow P(\limsup A_n) = 0 \Rightarrow S_n \xrightarrow{a.s} \mu.$$

Convergence in probability

Weak law of large numbers.

$S_n \longrightarrow \mu$ in probability sense

$$\lim_{n \to \infty} P(|S_n - \mu| > \varepsilon) = 0$$

$$A_n = \{\omega : |S_n(\omega) - \mu| > \varepsilon\}$$
$$= \{\omega : |\bar{S}_n(\omega)| > \varepsilon\}$$

$$P(A_n) = P_1(|S_n - \mu| > \varepsilon) \leq \frac{Var(S_n)}{\varepsilon^2}$$
$$= P((|\bar{S}_n|)^2 > \varepsilon^2) \leq \frac{E[\bar{S}_n^2]}{\varepsilon^2}$$

$$E[\bar{S}_n^2] = \frac{E[(\bar{X}_1 + \cdots \bar{X}_n)^2]}{n^2}$$

$$= \frac{\cancel{n} E[\bar{X}_1^2]}{n^{\cancel{2}}} = \frac{Var(X_1)}{n}$$

$$P(A_n) = P(|\bar{S}_n| > \varepsilon) = P(|S_n - \mu| > \varepsilon)$$

$$\leq \frac{Var(X_1)}{n}$$

$$0 \leq \lim_{n \to \infty} P(|S_n - \mu| > \varepsilon) \leq \lim_{n \to \infty} \frac{Var(X_1)}{n} = 0$$

$$\Rightarrow \lim_{n \to \infty} P(|S_n - \mu| > \varepsilon) = 0$$

Almost-sure convergence

$\Longrightarrow$ convergence in probability

$P(\limsup A_n) = 0$

$$\Longrightarrow \lim_{n\to\infty} P(A_n) = 0.$$

$$0 = P(\limsup A_n) = P\Big(\bigcap_{N\geq 1} \underbrace{\bigcup_{n\geq N} A_n}_{B_N}\Big)$$

$B_1 = \bigcup_{n=1} A_n$

$$= \lim_{N\to\infty} P(B_N)$$

$B_2 = \bigcup_{n=2} A_n$

$B_1 \supseteq B_2 \supseteq \cdots$

$$\geq \lim_{N\to\infty} P(A_N) \geq 0$$

non-negativity

$\boxed{A_N \subseteq B_N}$

$$\Rightarrow \lim_{N\to\infty} P(A_N) = 0.$$

Example:

$X_n =$

$X =$

X_n

$P($

$= P($

$Y = 2U$

$= P$

$=$

$= 0.$

$A_n)$

N

B_N

$B_N)$

$P(A_N) \geq 0$

non-negativity

$P(A_N) = 0.$

Example:

$X_n = u$ $\qquad u \sim \text{Uniform}[0,1]$

$X = 1-u$

$X_n \not\to X$ in probability sense

$P(|X_n - X| > \varepsilon)$

$= P(|2u-1| > \varepsilon)$

$Y = 2u-1 \sim \text{Uniform}[-1, 1]$

$= P(Y \notin [-\varepsilon, \varepsilon])$

$= 1 - \frac{2\varepsilon}{2} = 1-\varepsilon$

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Convergence of RVs.

Convergence in distribution sense.

$X_1, X_2, \ldots$ are said to converge to X in distribution sense if

$$\lim_{n\to\infty} F_{X_n}(x) = F_X(x). \quad \forall x.$$

Example: $X_n = U$, $X = 1-U$.

$F_{X_n}(x) = x \qquad x \in [0,1]$.

$F_X(x) = x. \qquad x \in [0,1]$.

$X_n \longrightarrow X$. in distribution sense.

Central Limit theorem

X_i's that are iid RVs with mean μ & variance σ^2

$$Z_n = \frac{\sum_{i=1}^{n}(X_i - \mu)}{\sqrt{n}\,\sigma} = \frac{\sum_{i=1}^{n}\bar{X}_i}{\sqrt{n}}$$

Let $\bar{X}_i = \frac{X_i - \mu}{\sigma}$

$\Rightarrow E[Z_n] = 0$, $Var(Z_n) = 1$

$Z_n \longrightarrow Z$ in distribution sense.

where $Z \sim N(0,1)$

We'll use inversion property of moment generating functions

$$\boxed{\lim_{n\to\infty} F_{Z_n}(z) = F_Z(z)}$$

$$\Updownarrow$$

$$\lim_{n\to\infty} \log M_{Z_n}(s) = \log M_Z(s)$$

$$M_{Z_n}(s) = E\left[e^{Z_n s}\right]$$

$$= E\left[e^{\frac{\sum_{i=1}^{n} \bar{X}_i s}{\sqrt{n}}}\right]$$

$$= E\left[\prod_{i=1}^{n} e^{\frac{\bar{X}_i s}{\sqrt{n}}}\right]$$

$\bar{X}_i$'s are identical

independent

$$= \prod_{i=1}^{n} E\left[e^{\frac{\bar{X}_i s}{\sqrt{n}}}\right] = \prod_{i=1}^{n} M_{\bar{X}_i}\left(\frac{s}{\sqrt{n}}\right) = \left(M_{\bar{X}_1}\left(\frac{s}{\sqrt{n}}\right)\right)^n$$

ons

$$\log M_{Z_n}(s)$$

$$= n \log M_{\overline{X}_1}\left(\frac{s}{\sqrt{n}}\right)$$

$$\lim_{n\to\infty} \log M_{Z_n}(s)$$

$$= \lim_{n\to\infty} \frac{\left(\log M_{\overline{X}_1}\left(\frac{s}{\sqrt{n}}\right)\right)}{\left(\frac{1}{n}\right)}$$

$$\boxed{L(s) = \log M_{\overline{X}_1}(s)}$$

'is are

ential

$$\left(M_{\overline{X}_1}\left(\frac{s}{\sqrt{n}}\right)\right)^n$$